	Analysis equation	Synthesis equation	Periodic?
CTFS	$a_k = \frac{1}{T_0}\int_{T_0} x(t)e^{-jkw_0t}\mathrm{d}t$	$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$	Not necessarily
DTFS	$a_k = \frac{1}{N_0}\sum_{n=\langle N_0\rangle} x[n]e^{-jk\Omega_0 n}$	$x[n] = \sum_{k=\langle N_0\rangle} a_k e^{jk\Omega_0 n}$	$a_k = a_{k+N_0}$
CTFT	$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\mathrm{d}t$	$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}\mathrm{d}w$	Not necessarily
DTFT	$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$	$x[n] = \frac{1}{2\pi}\int_{2\pi} X(j\Omega)e^{j\Omega n}\mathrm{d}\Omega$	$X(e^{j\Omega}) = X(e^{j(\Omega+2\pi)})$

Relationship between FS & FT analysis:

$$X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(w - k\omega_0).$$

If there is an aperiodic signal $g(t)$, how to compute its FS?

Replicate $g(t)$ in a period of T_0 (T_0 is not the length of $g(t)$), then

$$a_k = \frac{1}{T_0}\int_{T_0} g(t)e^{-jkw_0t}\mathrm{d}t.$$

Consider its FT:

$$G(j\omega) = \int_{-\infty}^{\infty} g(t)e^{-j\omega t}\mathrm{d}t.$$

So

$$\boxed{a_k = \frac{G(jk\omega_0)}{T_0}}.$$

which means a_k is sampling on $G(j\omega)$ with k then divided by T_0.

When the sampling is denser, the a_k approximates $G(j\omega)$ more, then $T_0 \to \infty$ more.

If $x(t)$ is real (i.e. $x(t) = x^*(t)$), then

$$X(j\omega) = X^*(-j\omega).$$

$$\boxed{\mathcal{E}v\{x(t)\} \overset{\mathscr{F}}{\longleftrightarrow} \mathcal{R}e\{X(j\omega)\}}.$$

$$\boxed{\mathcal{O}d\{x(t)\} \overset{\mathscr{F}}{\longleftrightarrow} j\mathcal{I}m\{X(j\omega)\}}.$$